

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Title: AI-Powered FAQ Chatbot for MPOnline Services

1. Introduction

1.1 Purpose

The purpose of this document is to describe the functional and non-functional requirements of the AI-powered FAQ chatbot system developed for MPOnline services. This system aims to provide instant and accurate responses to user queries using artificial intelligence.

1.2 Document Conventions

- The term “system” refers to the chatbot application
- “User” refers to any person interacting with the chatbot
- “Admin” refers to system administrator
- All requirements are listed in structured format

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Developers
- Project evaluators
- Organization stakeholders (MPOnline)

Readers are advised to start from Introduction and proceed sequentially.

1.4 Project Scope

The system provides an AI-based chatbot that allows users to ask questions related to MPOnline services. It also supports document-based knowledge retrieval, grievance submission, expert escalation, and analytics.

1.5 References

- FastAPI Documentation
- React Documentation

- PostgreSQL Documentation
- FAISS Documentation
- OpenAI / Gemini API Documentation

2. Overall Description

2.1 Product Perspective

The system is a web-based application consisting of frontend and backend components. It interacts with a database and AI models to generate responses.

2.2 Product Features

- User authentication system
- Public chatbot access
- AI-based query response
- Document upload with OCR
- Chat history management
- Grievance system
- Expert escalation
- Feedback system
- Analytics dashboard

2.3 User Classes and Characteristics

User

- Can ask questions
- Can submit grievances

Admin

- Manages documents
- Views analytics

Expert

- Handles escalated queries

2.4 Operating Environment

- Web browser (Chrome, Edge)
- Backend server (FastAPI)
- Database (PostgreSQL)
- OS (Windows/Linux)

2.5 Design and Implementation Constraints

- Requires internet connectivity
- Depends on AI API availability
- Limited by system resources

2.6 Assumptions and Dependencies

- Users have basic internet knowledge
- Documents are uploaded correctly
- API services are available

3. System Features

3.1 Functional Requirements

Authentication

- System shall allow user registration and login
- System shall validate credentials

Chatbot

- System shall accept user queries
- System shall generate AI-based responses
- System shall support multiple languages

Document Processing

- Admin shall upload documents
- System shall extract text using OCR
- System shall store embeddings

Grievance System

- User shall submit complaints
- System shall track status

Expert Escalation

- System shall forward unanswered queries
- Expert shall respond

Feedback

- User shall rate chatbot responses

Analytics

- System shall track usage
- Admin shall view reports

4. External Interface Requirements

4.1 User Interfaces

- Web-based UI using React
- Chat interface with input/output display
- Dashboard for admin

4.2 Hardware Interfaces

- System runs on standard computer/laptop
- No special hardware required

4.3 Software Interfaces

- Backend API (FastAPI)
- Database (PostgreSQL)
- AI APIs (OpenAI / Gemini)
- OCR (Tesseract)

4.4 Communications Interfaces

- HTTP/HTTPS protocol
- REST APIs

5. Nonfunctional Requirements

5.1 Performance Requirements

- System should respond within few seconds
- Should handle multiple users

5.2 Safety Requirements

- System should prevent unauthorized access
- Data should not be lost

5.3 Security Requirements

- Use JWT authentication
- Password hashing
- Role-based access

5.4 Software Quality Attributes

- Reliability
- Scalability
- Maintainability
- Usability